

Problem-Solution Fit

Date	30 June 2026
Team ID	
Project Name	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT Farmers Agricultural Researchers Policymakers Agricultural Officers Agribusiness Organizations	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES Limited internet access Budget constraints Limited digital literacy Lack of expert guidance Limited awareness of AI farming tools	5. AVAILABLE SOLUTIONS Traditional farming methods Expert consultation Basic agriculture websites PROS Easy to use Low cost CONS Less accurate No AI-based prediction Time-consuming Limited personalization

2. PROBLEMS / PAINS + ITS FREQUENCY Incorrect crop selection Low crop productivity Poor soil resource utilization Lack of scientific recommendations Frequent crop losses due to climate conditions	9. PROBLEM ROOT / CAUSE Lack of data-driven decisions Insufficient soil analysis Unpredictable weather conditions Traditional farming practices Limited access to personalized recommendations	7. BEHAVIOR + ITS INTENSITY Depends on farming experience Relies on local advice Uses weather information Slowly adopting digital farming technologies
3. TRIGGERS TO ACT Need for higher crop yield Climate change Rising farming costs Demand for smart farming Sustainable agriculture initiatives 4. EMOTIONS BEFORE Confused Uncertain Worried AFTER Confident Informed Satisfied	10. YOUR SOLUTION OptiCrop is an AI-powered web application that recommends the most suitable crop based on soil and environmental conditions. It enables farmers to make data-driven decisions, improve crop productivity, and promote sustainable agriculture.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Agriculture Websites Government Portals Mobile Apps YouTube Social Media OFFLINE Agricultural Extension centers Farmer Associations Local Markets Agricultural Training Programs